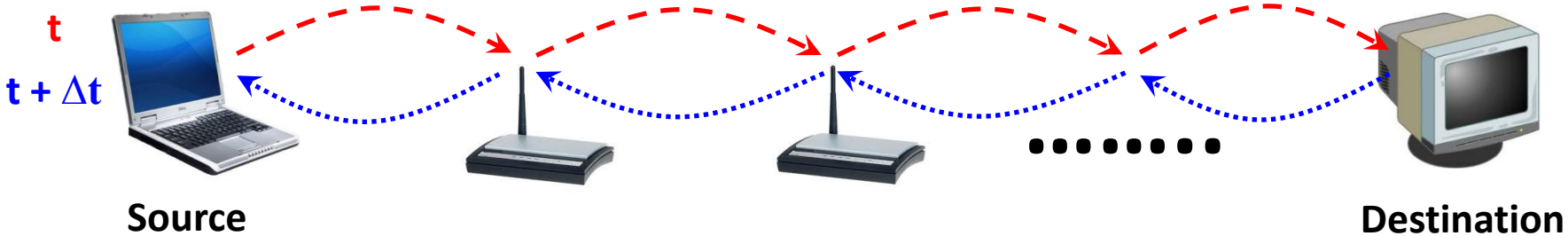
Your project could be...

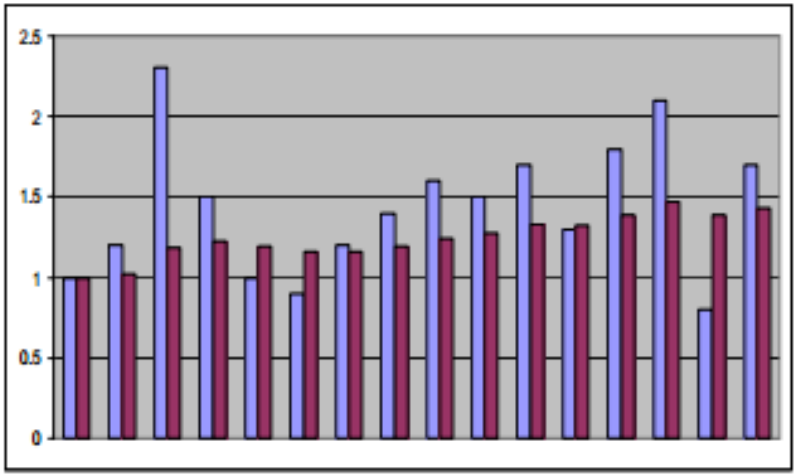
- Modified mechanism for RTT calculation
- Modified mechanism for congestion control in TCP
- Modified mechanism of routing
- Modified MAC layer protocol

RTT Calculation



$$RTT = \Delta t$$

- Round trip times exponentially averaged:
  - New RTT = α (old RTT) + (1 - α) (new sample)
  - Recommended value for α: 0.8 - 0.9
    - 0.875 for most TCP's



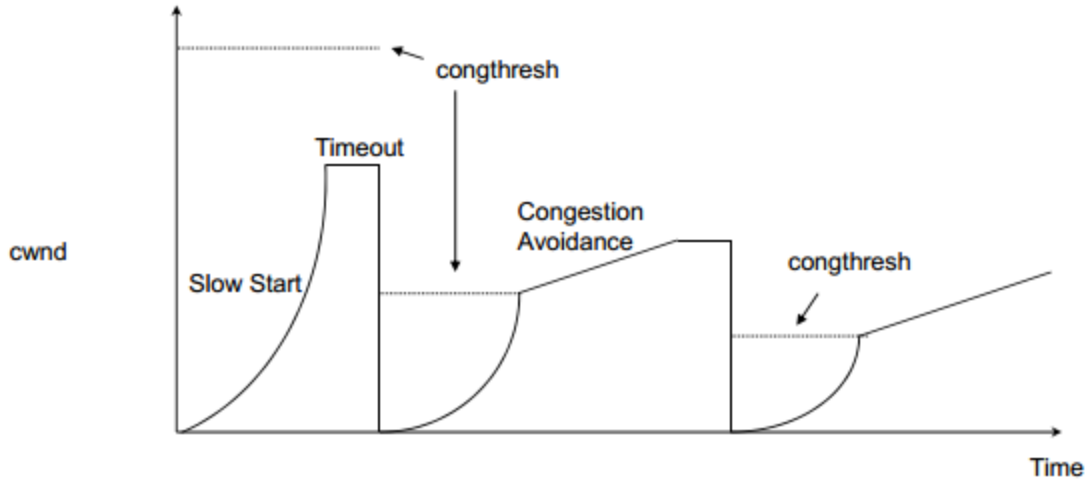
- Retransmit timer set to (b * RTT), where b = 2
  - Every time timer expires, RTO exponentially backed-off

Different Methods of RTT Calculations

- Jacobson
- Karn et al.
- Fallon et al.
- QRTT
- So on …

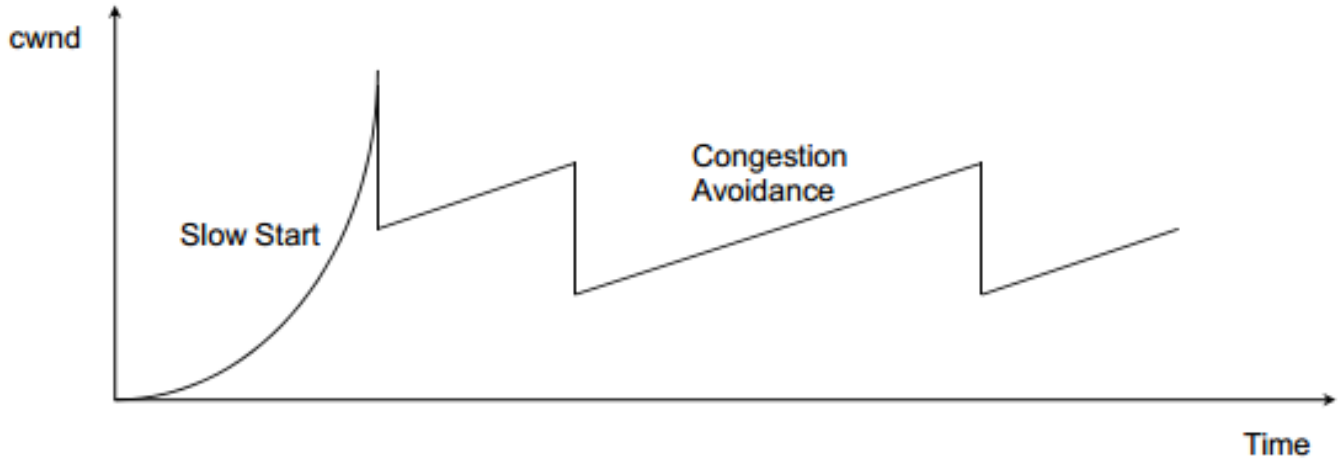
Congestion Control in TCP

- TCP Tahoe
  - Congestion Window (cwnd): Initial value is 1 MSS (=maximum segment size) counted as bytes
  - Slow-start threshold Value (CongestionThreshold = congthresh)
  - slow start (cwnd < congthresh)
    - Exponential increase of cwnd
  - congestion avoidance (cwnd >= congthresh)
    - Incremental increase of cwnd



Other Alternative of Congestion Control in TCP – TCP Reno

- Fast Retransmit and Fast Recovery



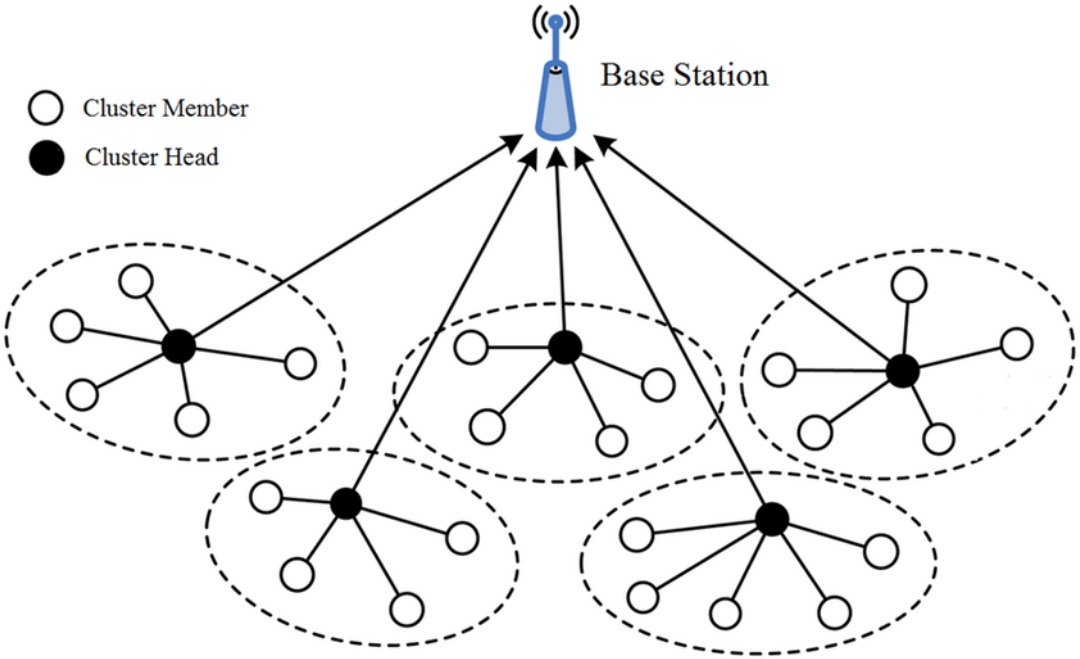
- Retransmit after 3 duplicate ACKs
  - prevent expensive timeouts
- Slow start only once per session (if no timeouts)
- In steady state, *cwnd* oscillates around the ideal window size.

Other Alternative of Congestion Control in TCP

- TCP NewReno
- TCP Vegas
- iTCP
- So on …

Routing Alternatives

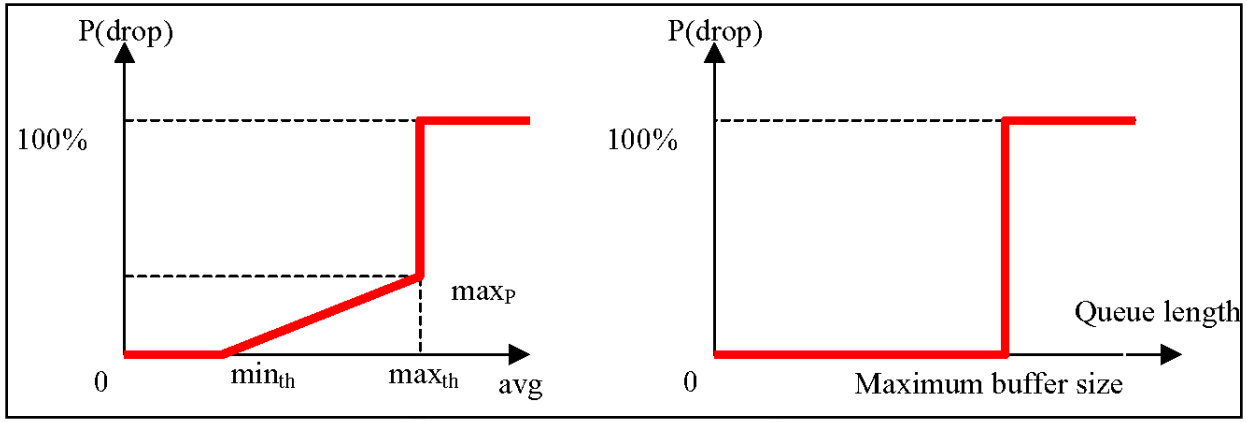
- Known routing techniques
  - DVR: IGRP
  - LSR: OSPF
- Other ones
  - BGP
  - Clustering

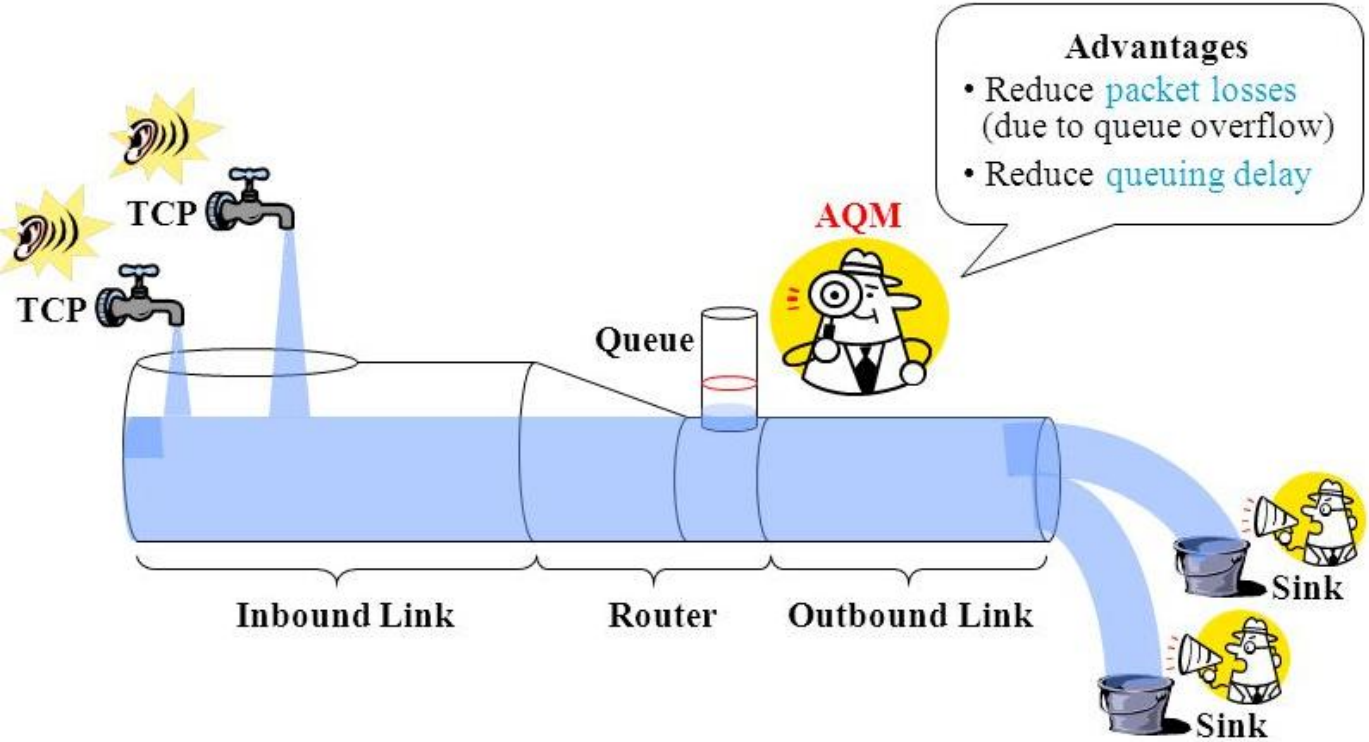


MAC Alternatives

- Known techniques
  - 802.11
  - 802.15.4
- Other alternatives
  - B-MAC from Berkeley
  - D-MAC (Dynamic or Directional)

Active Queue Management





5G and 6G

Features	5G	6G
Frequency Bands	• Sub 6 GHz, • mmwave for fixed access	• Sub 6 GHz, • mmwave for mobile access exploration of THz bands (>140 GHz), • Non-RF bands (e.g. optical, VLC) etc.
Data rate	1 Gbps to 20 Gbps (Downlink Data Rate - 20 Gbps, Uplink Data Rate - 10 Gbps)	1 Tbps
Latency (End to End Delay)	5 ms	< 1 ms
Traffic Capacity	10 Mbps/m^2	1 to 10 Gbps/m^2
Reliability	10^{-5}	10^{-9}
Localization precision	10 cm on 2D	1 cm on 3D
User experience	50 Mbps 2D everywhere	10 Gbps 3D everywhere

Some Resources …

- RTT, CC, Routing, and MAC
  - https://drive.google.com/drive/folders/17OfpS2kklbUTW5ZGDgrojlLI6Vgg8sBz?usp=sharing
- AQM
  - https://drive.google.com/file/d/1oEiv5RiH3tEwjVyiYNFnF-hj6I54kMSY/view
- 5G and 6G
  - https://www.rfwireless-world.com/Terminology/Difference-between-5G-and-6G.html
  - https://apps.nsnam.org/app/nr/
- Other sources of ns-3 based projects:
  - https://www.academia.edu/Documents/in/Ns-3_Simulator
  - https://www.nsnam.org/research/wns3/wns3-2016/accepted-papers/

Guidance and Caution …

- Try to do the tweaking work *COMPLETELY* at your own
  - Improvement in performance is not a requirement
  - Your (understanding + work) are the requirements

- Any identification of copying/cheating (from any source – web, classmate, or else) at any stage of evaluation: Double negative marking

Thank You

Questions are Welcome …